$f(x) = x^3 - 2x^2 - 4x + 8$

$f'(x) = 3x^2 - 4x - 4$

$f''(x) = 6x - 4$

$x_0 = 1.2$

Metodo estandar

$x_1 = f(1.2) = (1.2)^3 - 2(1.2)^2 - 4(1.2) + 8 = 2.048$

$f'(1.2) = 3(1.2)^2 - 4(1.2) - 4 = -4.48$

$x_1 = 1.2 - \left(\frac{2.048}{-4.48}\right) = 1.6571$

$e_a = \left|\frac{1.6571 - 1.2}{1.6571}\right| \times 100 = 27.58\%$

$x_2 = f(1.6571) = (1.6571)^3 - 2(1.6571)^2 - 4(1.6571) + 8$
$= 0.4287$

$f'(1.6571) = 3(1.6571)^2 - 4(1.6571) - 4 = -2.3855$

$x_2 = 1.6571 - \left(\frac{0.4287}{-2.3855}\right) = 1.8368$

$e_a = \left|\frac{1.8368 - 1.6571}{1.8368}\right| \times 100\% = 9.78\%$

$x_3 = f(1.8368) = (1.8368)^3 - 2(1.8368)^2 - 4(1.8368) + 8$
$= 0.0988$

$f'(1.8368) = 3(1.8368)^2 - 4(1.8368) - 4 = -1.2185$

$$X_3 = 1.8368 - \left(\frac{0.0488}{-1.2185}\right) = 1.9180$$

$$e_a = \left|\frac{1.9180 - 1.8368}{1.9180}\right| \times 100 = 4.23\%$$

$$X_4 = f(1.9180) = (1.9180)^3 - 2(1.9180)^2 - 4(1.9180) + 8$$
$$= 0.0238$$

$$f'(1.9180) = 3(1.9180)^2 - 4(1.9180) - 4 = -0.6120$$

$$X_4 = 1.9180 - \left(\frac{0.0238}{-0.6120}\right) = 1.9589$$

$$e_a = \left|\frac{1.9589 - 1.9180}{1.9589}\right| \times 100\% = 2.09\%$$

Y así hasta la iteración 13

$X_{13} = 1.999$ $\quad e_a = 0.001\%$

Metodo Newton Modificado.

$$X_1 = f'(1.2) = 6(1.2) - 4 = 3.2$$

$$X_2 = 1.2 - \frac{2.048 \cdot (-4.48)}{(-4.48)^2 - (2.048)(3.2)} = 1.8787$$

$$e_a = \left|\frac{1.8787 - 1.2}{1.8787}\right| \times 100\% = 36.12\%$$

$$V_2 = f(1.8787) = (1.8787)^3 - 2(1.8787)^2 - 4(1.8787) + 8$$
$$= 0.0450$$

$$f''(1.8787) = 3(1.8787)^2 - 4(1.8787) - 4 = -0.9250$$

$$F''(1.8787) = 6(1.8787) - 4 = 7.2721$$

$$x_2 = 1.8787 - \frac{(0.0450)(-0.9250)}{(-0.9250)^2 - (0.0450)(7.2721)}$$

$$x_2 = 1.9967$$

$$e_a = \left[\frac{1.9967 - 1.8787}{1.9967}\right] \times 100\% = 5.91\%$$

$$x_3 = F(1.9967) = (1.9967)^3 - 2(1.9967)^2 - 4(1.9967) + 8$$
$$= 0.000$$

$$F'(1.9967) = 3(1.9967)^2 - 4(1.9967) - 4 = -0.0264$$

$$F''(1.9967) = 6(1.9967) - 4 = 7.9802$$

$$x_3 = 1.9967 - \frac{(0.00)(-0.0264)}{(0.0264)^2 - (0.00)(7.98)}$$

$$x_3 = 2.000$$

$$e_a = 0\% \text{ ''}$$

el Metodo de Newton modificado necesito solo 3 iteraciones en esta fundión demostro ser mas eficiente que el Metodo estandar